

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“Spark Configuration Prediction System”

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

MANIKANTA B- 1DS20AI018

Conducted at

Rakuten

RAKUTEN

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

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CERTIFICATE

This is to certify that the Internship titled “**Spark Configuration Prediction System**” carried out by **Mr. MANIKANTA B[IDS20AI018]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering, in Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (**20AI8ICINT**).

Signature of Reporting Head
Pateriya, Saurabh
Senior Engineering Manager
RAKUTEN

Signature of Internship Coordinator
Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

Signature with Date

1) _____

2) _____

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **MANIKANTA B**, final year student of Information Science and Engineering, Dayananda Sagar College Of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **RAKUTEN**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date:

USN: 1DS20AI018

Place: Bangalore

NAME: MANIKANTA

OFFER LETTER



09-February-2024
Bangalore

Bukapindi Manikanta
DSATM

Internship Offer Letter

Dear Bukapindi,

Rakuten is pleased to offer you an internship opportunity as a "Technical Intern". You will report to Saurabh Pateriya. This position is located in Bangalore. You will begin your internship with us on 15-February-2024 and will be expected to work 5 days a week. The expected end date of your internship is 16-August-2024.

This letter is not a letter to offer employment but only an offer to Internship issued solely upon your request to enable you to complete your educational course. Further, please note that the stipend is paid to assist you in meeting the expenses of your education including college / tuition fees, purchase of books, subscriptions to portals with study materials, printing & stationery and such other expenses related to your education. The stipend amount is not in consideration of your services to Rakuten. Rakuten reserves its right to discontinue payment of stipend or the internship arrangement at any time with intimation to you.

You will be paid monthly stipend of INR 25000/- (Twenty Five Thousand only) including all applicable taxes if any. As an intern, you will receive benefits such as two meals served at company cafeteria, public transportation reimbursement (included in Stipend).

The Internship may be terminated at any time, either by you or Rakuten India without any reason by giving 30 calendar days' written notice or 30 days' stipend in lieu of the notice to the other party.

By accepting this offer, you agree that throughout your internship, you will observe all policies and practices governing the conduct of our business and employees.

I hope that your association with the Company will be successful and rewarding. Please indicate your acceptance of this offer by signing below and returning a copy of the same for Company records.

Sincerely,
For Rakuten India Enterprise Private Limited

Nalini George
Chief People Officer,
Rakuten India Enterprise Private Limited

I accept internship opportunity at Rakuten India on the terms and conditions set out in this letter.

B. Manikanta
13/02/24

Candidate Acceptance Sign and date

Rakuten India Enterprise Private Limited

Rakuten Crimson House

CIN - U72200KA2016FTCO85435

Phone: +91 636 686 6476 | Email: rakuten-contact@mail.rakuten.com

Regd. Office: #20, 1st cross, Raja Ram Mohan Roy Road, Sampangiramanagar, Bengaluru, Karnataka-560027

[Rakuten India Webpage](#)

CONFIDENTIAL

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Mr. Pateriya Saurabh**, Rakuten, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, **Prof. Kavya D N**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

MANIKANTA B

[1DS20AI018]

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CHAPTER 1

INTRODUCTION

Introduction to spark:

Spark resource configuration involves optimizing the allocation of computational resources within a Spark application to ensure efficient execution of tasks and optimal utilization of cluster resources. This involves specifying various parameters related to CPU, memory, and parallelism to meet the specific requirements of the application and the underlying infrastructure.

Key configurations include:

1. **Driver Memory:** The amount of memory allocated to the driver process, which coordinates the execution of the Spark application.
2. **Executor Memory:** The memory allocated to each executor, which is responsible for running tasks on individual worker nodes.
3. **Number of Executors:** Specifies the total number of executor instances to be launched on the cluster, influencing parallelism and resource utilization.
4. **Executor Cores:** Defines the number of CPU cores allocated to each executor, determining the degree of parallelism for task execution.

1. Sparklens:

Sparklens is a powerful tool designed to analyze and optimize Apache Spark applications, providing detailed insights into resource usage, performance bottlenecks, and cluster efficiency. By analyzing Spark application logs and metrics, Sparklens offers valuable recommendations for tuning various configurations such as executor memory, CPU cores, and parallelism settings to maximize application performance and resource utilization. Its comprehensive reports and visualizations help developers and administrators identify issues, optimize resource allocation, and improve the overall efficiency of Spark applications, ultimately leading to faster execution times and reduced costs in big data processing workflows.

2. Dr. Elephant:

Dr. Elephant is an open-source performance monitoring and tuning tool designed specifically

for Apache Hadoop and Apache Spark clusters. Developed by LinkedIn, Dr. Elephant aims to optimize the resource usage and performance of Hadoop and Spark jobs by identifying inefficiencies and suggesting optimizations. It analyzes job execution patterns, resource utilization, and configuration settings to provide insights into potential bottlenecks and areas for improvement. Dr. Elephant offers recommendations for tuning parameters such as memory allocation, CPU utilization, and parallelism settings to enhance job performance and resource efficiency. With its intuitive interface and actionable insights, Dr. Elephant helps organizations maximize the performance of their big data processing workloads running on Hadoop and cluster.

3. Kafka:

Apache Kafka is a distributed streaming platform designed to handle high-throughput, fault-tolerant, and real-time data streams. It serves as a centralized hub for building real-time data pipelines and applications, enabling seamless communication between systems and applications through its distributed messaging system. Kafka's architecture consists of topics, producers that publish data to topics, consumers that subscribe to topics, and brokers that manage the storage and replication of data across a cluster of servers. Its scalability, durability, and support for horizontal scaling make Kafka a popular choice for use cases such as log aggregation, event sourcing, stream processing, and real-time analytics in various industries, including finance, retail, and technology.

4. Spark History Server:

The Spark History Server is a component of Apache Spark that provides a web-based interface to view details about completed Spark applications. It collects and stores information about Spark jobs, stages, tasks, and execution metrics, allowing users to analyze performance, troubleshoot issues, and monitor the progress of past Spark applications. The History Server retrieves data from event logs generated by Spark applications and presents it in a user-friendly format through a web UI. Users can access information such as job duration, resource usage, task execution times, and job DAG (Directed Acyclic Graph) visualization. This enables developers, data engineers, and administrators to gain insights into the behavior of Spark applications over time, aiding in performance optimization and resource management.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

RAKUTEN



Introduction

Rakuten, which means “**optimism**” in Japanese, is a leading global company that empowers individuals, businesses, and societies to realize their dreams. Their philosophy of “**Walk Together**” expresses their commitment to building a better, more optimistic future by embracing new ideas and acting on them with operational agility.

Rakuten, Inc., a Japanese multinational conglomerate headquartered in Tokyo, stands as a global leader in e-commerce, fintech, digital content, communications, and more. Founded in 1997 by Hiroshi Mikitani, Rakuten has grown from an online marketplace to a diverse ecosystem of services, spanning industries and continents. With a commitment to innovation and customer-centricity, Rakuten offers a wide array of platforms and products, including Rakuten Ichiba, Rakuten Mobile, Rakuten Kobo, and Rakuten Viber, among others. Its presence extends beyond commerce, with investments in sports, entertainment, and technology sectors, marking Rakuten as a dynamic force shaping the digital landscape worldwide.

Founder



Hiroshi Mikitani, the visionary founder and CEO of Rakuten, Inc., has forged a remarkable path in the world of business, marked by innovation and global impact. Raised in Kobe, Japan, Mikitani's upbringing under the guidance of his father, Ryōichi Mikitani, instilled in him a deep appreciation for academia and economic principles. After graduating from Hitotsubashi University, Mikitani embarked on a career in banking before pursuing his entrepreneurial ambitions. Following his tenure at the Industrial Bank of Japan, he ventured into consulting and eventually founded Rakuten Ichiba in 1997, laying the foundation for what would become Rakuten, Inc. Under his leadership, Rakuten evolved into a powerhouse of e-commerce, embodying Mikitani's vision to empower the world through innovation. Beyond Rakuten, Mikitani's influence extends to various spheres; he presides over Crimson Group, chairs the football club Vissel Kobe and the Tokyo Philharmonic Orchestra, and contributes as a board member of Lyft. Mikitani's entrepreneurial journey serves as a beacon of inspiration for aspiring innovators worldwide, showcasing the transformative power of dedication, vision, and relentless pursuit of excellence.

ABOUT THE COMPANY

Rakuten, Inc. is a Japanese multinational conglomerate that operates a diverse array of internet services, spanning e-commerce, fintech, digital content, communications, and more. Founded in 1997 by Hiroshi Mikitani, Rakuten started as an online marketplace called Rakuten Ichiba, which quickly grew to become one of Japan's largest e-commerce platforms. Over the years, Rakuten expanded its reach both domestically and internationally through strategic acquisitions and partnerships.

VISION STATEMENT

At Rakuten, we drive disruptive innovation, Engaging knowledge, creativity and passion from around the world. To achieve ambitious goals and help build communities in which people can pursue their dreams and live in happiness.

MISSION STATEMENT

- Contribute to society by creating value through innovation and entrepreneurship
- Empowering people to realize their hopes and dreams
- Embracing new thinking
- Rakuten changes the world through innovation.

Internship - Positions available at Rakuten for the following Profiles when applied through campus placement cell is,

- Technical Intern.

CHAPTER 3

SCOPE OF WORK

PROBLEM STATEMENT

In data processing pipelines utilizing Apache Spark, the challenge lies in effectively configuring resources to accommodate datasets of varying sizes. It's imperative to strike a balance between optimizing costs and maintaining processing efficiency. This entails dynamically adjusting Spark resource allocations based on dataset sizes to avoid unnecessary expenses while ensuring timely and effective data processing.

Introduction to Spark Configuration Prediction System

In the realm of big data processing, Apache Spark stands out as a powerful framework for distributed computing. However, configuring Spark resources optimally presents a significant challenge, especially when dealing with datasets of varying sizes. This challenge becomes even more pronounced in the context of machine learning (ML) algorithms, where the computational demands fluctuate based on the complexity of the algorithm and the characteristics of the dataset.

To address this issue, there is a growing interest in leveraging ML algorithms to predict the optimized configuration for Spark resources. By harnessing the predictive capabilities of ML models, organizations can dynamically adjust Spark configurations in real-time, thereby maximizing cost-effectiveness while ensuring efficient processing of data.

This sets the stage for exploring how ML algorithms can revolutionize the optimization of Spark resource configurations, leading to enhanced performance and reduced operational costs in big data environments.

The burgeoning field of big data analytics has necessitated the development of robust and scalable computing frameworks, among which Apache Spark has emerged as a frontrunner. Spark's ability to process large volumes of data across distributed systems has made it a go-to solution for a wide range of data-intensive applications. However, the task of configuring Spark to efficiently handle diverse datasets and computational workloads remains a

formidable challenge. This challenge is particularly acute in the deployment of machine learning (ML) algorithms, where the optimal allocation of resources such as memory and CPU cores can vary significantly depending on the complexity of the algorithm and the characteristics of the data being processed.

In response to these challenges, there is a growing interest in the application of ML algorithms to predict and optimize Spark configurations. This innovative approach seeks to leverage the predictive power of ML to dynamically tailor Spark resource allocations to the specific needs of each task. By doing so, it aims to streamline the configuration process, enhance computational efficiency, and reduce the time and resources required to process big data.

The premise of using ML for Spark configuration optimization is rooted in the idea that historical data on Spark job performances can be analyzed to identify patterns and insights that inform future configurations. This involves training ML models on a variety of metrics, such as execution times, resource utilization rates, and job outcomes, under different configurations. These models can then predict the most effective configuration settings for new jobs, taking into account the unique demands of each task.

The potential benefits of this approach are significant. By automating the optimization of Spark configurations, organizations can achieve more efficient data processing, reduce operational costs, and allocate resources more effectively. This not only enhances the performance of Spark-based applications but also supports more scalable and cost-effective big data analytics practices.

As the intersection of ML and big data continues to evolve, the development of systems for predicting and optimizing Spark configurations represents a promising frontier. This initiative has the potential to transform big data processing by making it more adaptive, efficient, and accessible, thereby enabling organizations to harness the full power of their data assets.

CHAPTER 4

LEARNING OBJECTIVES

1. Mastery of Rakuten Ecosystem:

Interns participating in the Rakuten internship program will acquire a comprehensive understanding of the diverse Rakuten ecosystem, encompassing e-commerce, fintech, digital content, communications, and more. Through immersive training and guided learning modules, interns will explore the core components and services offered by Rakuten, including Rakuten Ichiba, Rakuten Mobile, Rakuten Kobo, and Rakuten Viber. By delving into the intricacies of Rakuten's business model and technology stack, interns will develop proficiency in navigating and leveraging the various platforms and products within the Rakuten ecosystem.

2. Digital Commerce and Customer Engagement:

Interns will immerse themselves in the realm of digital commerce and customer engagement, gaining insights into the strategies and technologies employed by Rakuten to drive customer acquisition, retention, and loyalty. From understanding consumer behavior and market trends to exploring innovative marketing techniques and personalized shopping experiences, interns will deepen their knowledge of digital commerce best practices. Additionally, interns will explore the role of data analytics, machine learning, and customer relationship management (CRM) tools in optimizing customer engagement and driving business growth within the Rakuten ecosystem.

3. Innovation and Entrepreneurship:

With Rakuten's commitment to innovation and entrepreneurship at its core, interns will have the opportunity to explore and contribute to cutting-edge projects and initiatives within the company. Through hands-on projects, collaborative brainstorming sessions, and mentorship from industry experts, interns will cultivate their entrepreneurial mindset and creativity, identifying opportunities for innovation and disruption in the digital marketplace. By immersing themselves in Rakuten's culture of innovation, interns will develop the skills and mindset necessary to thrive in dynamic and fast-paced environments, laying the foundation for future success in the tech, fintech and communication(Netwrok) industries and beyond.

4. Technical Skill Development:

Interns at Rakuten will not only gain insights into business and marketing strategies but will also have the opportunity to enhance their technical skills. The program is designed to offer a deep dive into the latest technologies and software development practices that power the Rakuten ecosystem. From learning about cloud computing services that support Rakuten's vast online platforms to understanding the intricacies of cybersecurity measures that protect user data, interns will acquire hands-on experience in critical technical domains. Additionally, the program will introduce interns to the world of data science and analytics, enabling them to leverage big data to make informed decisions and drive innovation within the company. By the end of their internship, participants will have developed a strong foundation in the technical skills required to excel in the digital economy.

5. Global Networking and Cultural Exchange:

Rakuten's global presence offers interns a unique opportunity to engage in cross-cultural collaboration and networking. Interns will interact with professionals and fellow interns from around the world, gaining diverse perspectives on technology, business, and innovation. This aspect of the program is designed to enhance interns' global awareness and communication skills, preparing them for successful careers in an increasingly interconnected world. Through workshops, team-building activities, and informal gatherings, interns will build a valuable network of contacts and develop friendships that span the globe.

6. Professional Development and Career Guidance:

The Rakuten internship program places a strong emphasis on the professional development of its participants. Interns will receive personalized career guidance, including resume workshops, interview preparation sessions, and insights into career paths within Rakuten and the broader tech industry. Mentorship from experienced professionals will provide interns with guidance and feedback on their performance and career aspirations. This supportive environment aims to equip interns with the knowledge, skills, and confidence to pursue their professional goals, whether they aspire to join Rakuten upon completion of their internship or explore opportunities elsewhere in the tech ecosystem.

By participating in the Rakuten internship program, interns will emerge with a well-rounded experience, having developed a deep understanding of the Rakuten ecosystem, honed their technical and professional skills, and built a global network of contacts. This comprehensive approach ensures that interns are not only prepared to meet the challenges of the digital economy but are also positioned to become future leaders in technology and innovation.

CHAPTER 5

CHALLENGES AND SOULTION

In the development of a Spark configuration prediction system aimed at optimizing job performance and resource utilization, various non-functional challenges must be considered alongside the system's functionalities. While meeting user requirements is paramount, ensuring accuracy, scalability, maintainability, reliability, and security are essential aspects of software engineering.

a) Conformance to specific standards: The Spark configuration prediction system must adhere to industry standards and best practices in distributed computing and machine learning to ensure compatibility and interoperability with existing Spark frameworks and environments.

b) Performance constraints: The system should be optimized to handle large-scale datasets and complex prediction algorithms efficiently, without sacrificing response times or system throughput.

c) Hardware limitations: Consideration must be given to the hardware resources available for running the Spark configuration prediction system, ensuring compatibility and scalability across different cluster configurations and computing infrastructures.

d) Maintainability: The system's codebase should be well-structured and modular, facilitating easy maintenance, updates, and enhancements over time as Spark evolves and new optimization techniques emerge.

e) Reliability: The system must be robust and fault-tolerant, capable of handling errors, failures, and unexpected inputs gracefully, while maintaining data integrity and system availability during prediction tasks.

f) Testability: Adequate testing mechanisms should be in place to validate the accuracy and reliability of the Spark configuration prediction system's predictions, including unit tests, integration tests, and performance testing scenarios.

Solution:

To address accuracy challenges in predicting optimal Spark configurations, leveraging machine learning techniques such as regression analysis and ensemble methods offers a promising solution. By training predictive models on historical job performance data and Spark configuration parameters, the system can learn patterns and correlations to accurately predict optimal configurations for future Spark jobs. Additionally, incorporating techniques like cross-validation and hyperparameter tuning can further enhance the accuracy and robustness of the prediction models. Integrating these machine learning models into the Spark configuration prediction system's architecture can significantly improve its accuracy and reliability, thereby addressing one of the key challenges in its development in current real world scenarios.

HARDWARE AND SOFTWARE REQUIREMENTS**Software requirements:**

- Front End: Streamlit
- Server: Spark History Server

Hardware Requirements:

- Processor: Intel core i5 processor.
- Memory: 15.6 GB.
- Hard Disk: 40 GB.

CHAPTER 6

ACHEIVEMENTS AND CONTRIBUTION

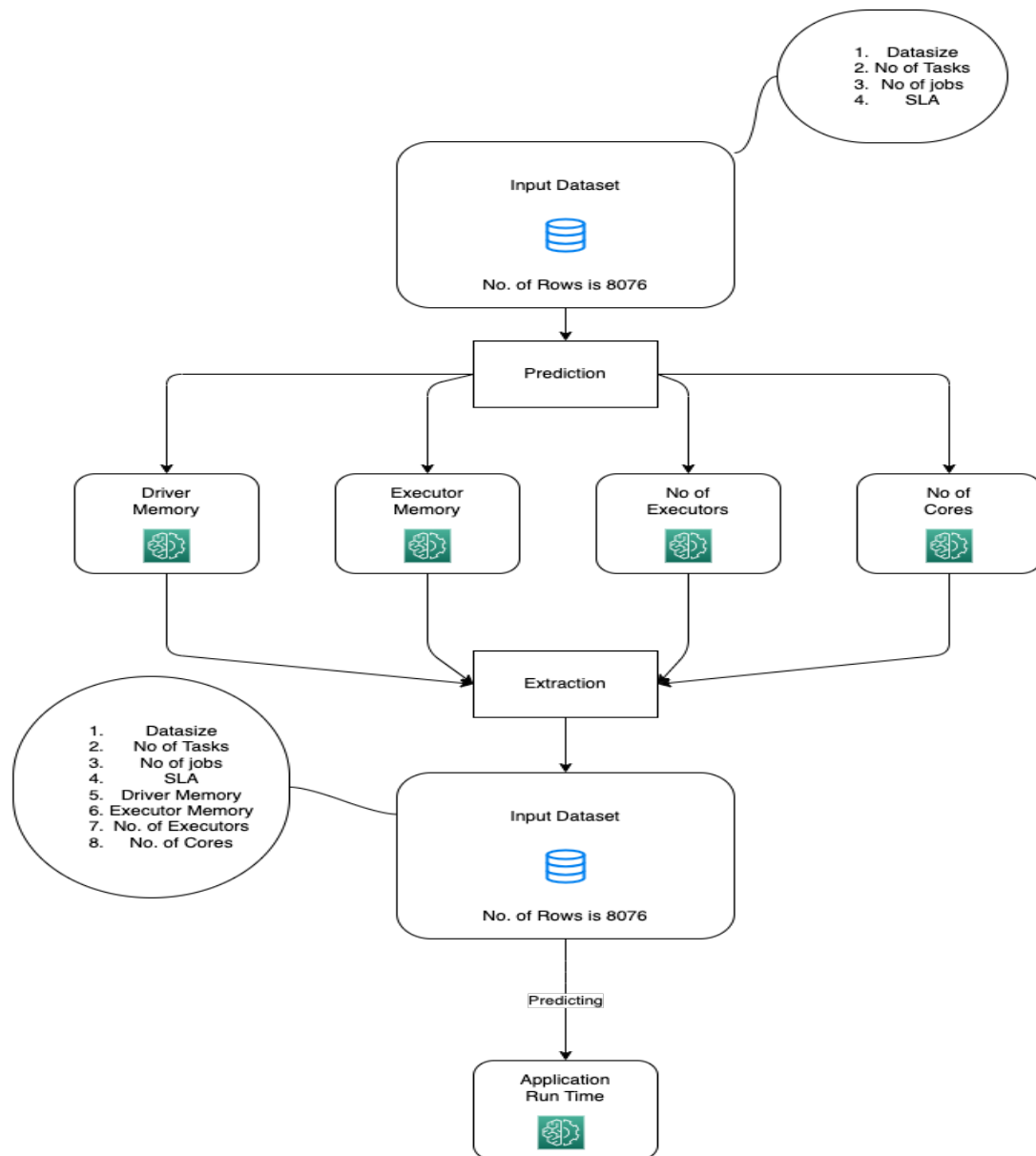


Fig: System Workflow

The System mainly contains:

- **Prediction:** In our approach to optimizing Spark resource configuration, we employ the XGBoost algorithm for predictive modeling. XGBoost is chosen for its robust handling of outliers, a common challenge in resource allocation tasks. We focus on predicting key parameters such as driver memory, executor memory, number of executors, and number of cores using XGBoost. By leveraging XGBoost's capability to effectively manage outliers, we aim to enhance the accuracy and reliability of our predictions, leading to

more efficient resource utilization in Apache Spark environments.

- Extraction: After predicting the optimized Spark resource configuration using XGBoost, the next step involves leveraging these predicted resources to forecast the application runtime. By incorporating the anticipated driver memory, executor memory, number of executors, and number of cores into our model, we utilize machine learning techniques to estimate the runtime of Spark applications. This process enables us to proactively assess the performance impact of the predicted resource configuration, facilitating informed decision-making and further optimization of data processing workflows.
- Spark Configuration Prediction System: The Spark Configuration Prediction System is a sophisticated tool engineered to optimize the performance and resource utilization of Spark applications, providing users with predictive insights into optimal configuration parameters. Harnessing machine learning algorithms and historical job performance data, the system analyzes the characteristics of Spark jobs and recommends configuration settings to maximize efficiency and throughput. By automating the process of tuning Spark parameters, it empowers users to achieve optimal performance without the need for manual experimentation. Additionally, the system offers insights into potential bottlenecks, scalability challenges, and resource constraints, enabling users to proactively address issues and enhance the overall efficiency of their Spark deployments. Through its predictive capabilities, the Spark Configuration Prediction System revolutionizes the management of Spark clusters, driving productivity and facilitating informed decision-making in big data processing environments.

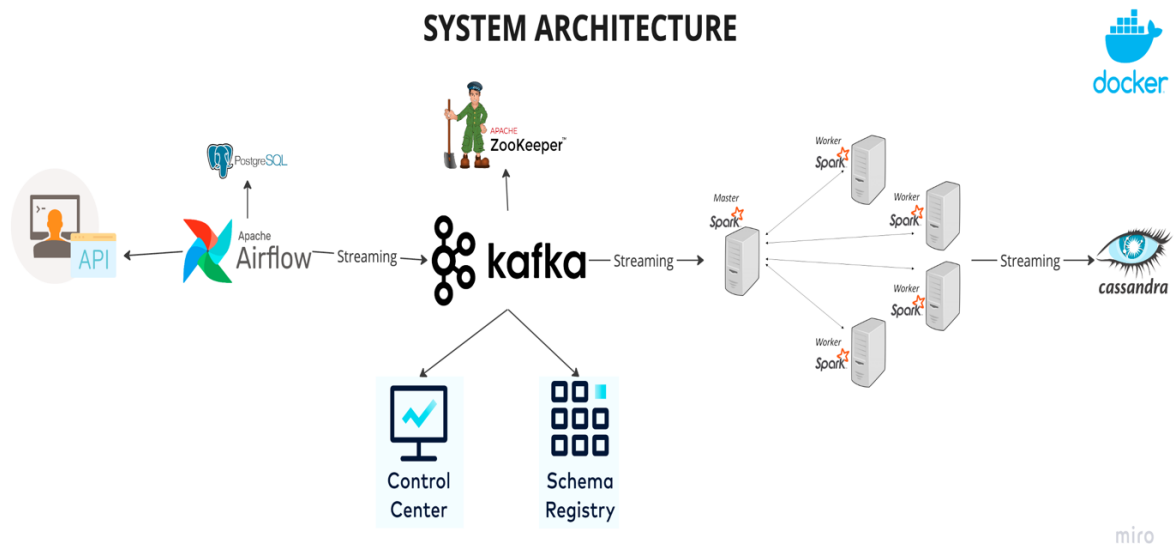


Fig: System Design

Let's consider a generic data processing and workflow management system as described in the previously analyzed system architecture diagram. This will help in understanding how different components interact within a workflow to process data from ingestion to storage and analysis.

1. Data Ingestion:

API Gateway: The process begins with data ingestion through an API Gateway. External clients submit data to the system via this API, which acts as the entry point. This ensures that data enters the system in a controlled and secure manner.

2. Workflow Orchestration:

Apache Airflow: Once the data is ingested, Apache Airflow takes over for workflow orchestration. Airflow schedules and manages the execution of data processing tasks. It ensures that tasks are executed based on their dependencies and predefined schedules. This could involve data transformation, validation, or any other processing logic required by the system.

3. Data Streaming and Processing:

Apache Kafka: For real-time data processing needs, the system uses Apache Kafka. Kafka

streams the data across different parts of the system, enabling high-throughput and fault-tolerant messaging. This is crucial for scenarios where data needs to be processed in real-time or near-real-time.

4. Distributed Coordination:

Zookeeper: Apache ZooKeeper provides distributed coordination services for Kafka. It helps in managing the Kafka cluster, ensuring that the system maintains high availability and fault tolerance. ZooKeeper keeps track of the status of Kafka brokers, partitions, and ensures consistent configuration across the cluster.

5. Data Storage:

PostgreSQL Database: After processing, the data is stored in a PostgreSQL database. This relational database system stores processed data, making it available for querying, analysis, and reporting. It can also store metadata for Airflow workflows, enabling efficient management and monitoring of data processing jobs.

6. Monitoring and Management:

Control Center: A Control Center or a similar dashboard provides a user interface for monitoring and managing the system. It offers insights into the system's performance, data flow, and operational metrics. This enables system administrators and data engineers to keep track of the system's health and make informed decisions.

Workflow Summary:

The complete process in this workflow involves data ingestion through an API, orchestration of data processing tasks with Apache Airflow, real-time data streaming with Apache Kafka, coordination with ZooKeeper, data storage in PostgreSQL, and system monitoring through a Control Center. This workflow exemplifies a robust system architecture designed for efficient data processing and workflow management, ensuring data is accurately processed, stored, and made ready for analysis and decision-making.

Flow of the process

Histroy events > CSV > XGBoost > Configuration Prediction

During my internship, I spearheaded the development of an advanced AI Assistant tailored specifically for reporting tasks within the organization. This Spark Configuration Prediction represented a significant milestone in enhancing reporting efficiency and accuracy, leveraging cutting-edge technologies in natural language processing and data analytics. By harnessing the power of machine learning algorithms and advanced language models, I engineered a solution capable of automating report generation, analyzing complex datasets, and extracting actionable insights from large volumes of data in the spark configuration.

CHAPTER 7

REFLECTION AND ANALYSIS

My internship experience at Rakuten provided me with a rich and rewarding opportunity to explore the intersection of technology and business within one of the world's leading internet services companies. Working in Rakuten's innovative environment allowed me to gain invaluable insights into key concepts and technologies, particularly in the realm of Spark configuration prediction and optimization.

Throughout my internship, I had the privilege of delving deep into Spark configuration prediction, a critical component of Rakuten's data processing infrastructure. By leveraging machine learning algorithms and historical job performance data, I learned how to analyze and optimize Spark configurations to enhance performance and resource utilization. Collaborating with experienced data engineers and analysts, I gained hands-on experience in fine-tuning Spark parameters, identifying bottlenecks, and maximizing the efficiency of Spark clusters.

Moreover, my internship at Rakuten provided me with exposure to the broader ecosystem of internet services and digital commerce. By immersing myself in Rakuten's diverse range of products and platforms, including Rakuten Ichiba, Rakuten Mobile, and Rakuten Kobo, I gained a holistic understanding of Rakuten's business model and technology stack. This exposure not only broadened my perspective but also deepened my appreciation for the role of technology in driving innovation and business growth.

Furthermore, my internship experience at Rakuten emphasized the importance of collaboration, innovation, and continuous learning. Working alongside talented professionals from diverse backgrounds, I was inspired by Rakuten's culture of creativity and entrepreneurship. Through collaborative projects, mentorship, and knowledge-sharing sessions, I was able to expand my skill set, refine my problem-solving abilities, and gain practical insights into real-world challenges.

In reflection, my internship at Rakuten was a transformative journey that provided me with invaluable opportunities for personal and professional growth. From mastering Spark configuration prediction to understanding the intricacies of digital commerce, I emerged from

my internship experience equipped with new skills, insights, and perspectives. Moving forward, I am excited to apply what I have learned to contribute to Rakuten's ongoing success and to continue my journey of learning and exploration in the dynamic world of technology.

CHAPTER 8

RECOMMENDATIONS

Flexible Training Schedule:

Offer interns the flexibility to choose their working hours, accommodating their individual preferences and schedules. By allowing interns to work during times that suit them best, you empower them to maintain a healthy work-life balance and optimize their productivity.

Structured Training Modules:

Develop structured training modules and resources to guide interns through the onboarding process and provide them with a clear roadmap for skill development. These modules should cover essential topics related to the company's technologies, processes, and culture, ensuring that interns receive comprehensive training that aligns with their learning objectives.

Mentorship Program:

Implement a mentorship program that pairs each intern with an experienced mentor who can provide guidance, feedback, and support throughout their internship. Mentors play a crucial role in facilitating learning, fostering professional growth, and helping interns navigate challenges effectively. Encourage mentors to schedule regular check-ins with their mentees and provide ongoing support and encouragement.

Hands-on Project Assignments:

Assign interns to hands-on project assignments that allow them to apply their knowledge and skills in real-world scenarios. These projects should be challenging yet achievable, providing interns with opportunities to learn new concepts, solve complex problems, and demonstrate their capabilities. Encourage interns to collaborate with their peers and mentors to tackle project tasks collaboratively and share their insights and learnings.

Feedback and Evaluation:

Provide interns with regular feedback and evaluation sessions to assess their progress, identify areas for improvement, and recognize their achievements. Feedback sessions should be constructive, specific, and actionable, highlighting interns' strengths and areas for development. Encourage interns to reflect on their experiences, set goals for their learning journey, and take ownership of their professional development. By fostering a culture of continuous feedback and improvement, you empower interns to maximize their potential and make meaningful contributions to the company.

CHAPTER 9

CONCLUSION

1. **High Predictive Accuracy:** Our predictive models, leveraging XGBoost, have demonstrated impressive performance, achieving R-squared values exceeding 90% for each predicted Spark resource configuration. Additionally, the overall application runtime model exhibits a commendable accuracy of 84%.
2. **Need for Individual App Analysis:** Despite the success of our predictive models, it's essential to acknowledge the challenge posed by imbalanced data. To ensure robustness and reliability across various Spark applications, further analysis of individual applications is warranted. This granular examination will help identify and address specific performance nuances, enabling tailored optimization strategies for each application scenario.
3. **Continuous Improvement:** Moving forward, a commitment to continuous improvement remains paramount. By iteratively refining our models and methodologies, we can enhance predictive accuracy, mitigate data imbalances, and ultimately optimize Spark resource configuration and application runtime across diverse usage scenarios.

CHAPTER 10

REFERENCES

[1] Founder : Mickey Mikitani

LinkedIn : <https://jp.linkedin.com/in/mikitani>

[2] Team Lead : Vimal Chaudhari

LinkedIn : <https://in.linkedin.com/in/vimal-chaudhari>

[3] Apache Zookeeper : <https://zookeeper.apache.org/>

[4] Rakuten : <https://global.rakuten.com/corp/>

[5] Apache Spark : <https://spark.apache.org/>

[6] History Server : <https://docs.datastax.com/en/dse/5.1/docs/spark/configure-history->

[7] Apache Kafka : <https://kafka.apache.org/>

[8] Streamlit : <https://streamlit.io/>

[9] Beautiful Soup : <https://beautiful-soup-4>.

[10] Apache Airflow : <https://airflow.apache.org/>